Math 550, S20
Quiz 5

Name: Answer

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Let $\mathbf{F} = x^2\mathbf{i} - xe^{2y}\mathbf{j} + 3xyz\mathbf{k}$.

(a) Compute the divergence of $\mathbf{F}$, $\nabla \cdot \mathbf{F}$.

$$\begin{aligned}\nabla \cdot F &= \nabla \cdot (x^2, -xe^{2y}, 3xyz) \\ &= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(-xe^{2y}) + \frac{\partial}{\partial z}(3xyz) \\ &= 2x - 2xe^{2y} + 3xy \\ &= x(2 - 2e^{2y} + 3y)\end{aligned}$$

(b) Compute the curl of $\mathbf{F}$, $\nabla \times \mathbf{F}$.

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & -xe^{2y} & 3xyz \end{vmatrix} = i\begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -xe^{2y} & 3xyz \end{vmatrix} - j\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ x^2 & 3xyz \end{vmatrix} + k\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ x^2 & -xe^{2y} \end{vmatrix}$$

$$= i(3xz - 0) - j(3yz - 0) + k(-e^{2y} - 0)$$

$$= 3xz\,i - 3yz\,j - e^{2y}k$$

(c) Compute $\nabla \cdot (\nabla \times \mathbf{F})$ based on the result from (b).

$$\begin{aligned}\nabla \cdot (\nabla \times F) &= \nabla \cdot (3xz, -3yz, -e^{2y}) \\ &= \frac{\partial}{\partial x}(3xz) + \frac{\partial}{\partial y}(-3yz) + \frac{\partial}{\partial z}(-e^{2y}) \\ &= 3z - 3z + 0 = 0\end{aligned}$$

Problem 2. (4 points) Find the first and second order Taylor polynomials, $P_1(x, y)$ and $P_2(x, y)$, for the function $f(x, y) = e^{2x+y}$ at the point $\mathbf{a} = (0, 0)$.

$$Df = [\underset{f_x}{2e^{2x+y}} \quad \underset{f_y}{e^{2x+y}}] \Rightarrow Df(0,0) = [2 \quad 1]$$

$$P_1(x,y) = f(0,0) + \underbrace{Df(0,0)\begin{bmatrix} x-0 \\ y-0 \end{bmatrix}} \Leftarrow f_x(0,0)(x-0) + f_y(0,0)(y-0)$$

$$= 1 + [2 \quad 1]\begin{bmatrix} x \\ y \end{bmatrix}$$

$$= 1 + 2x + y$$

$$Hf = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 4e^{2x+y} & 2e^{2x+y} \\ 2e^{2x+y} & e^{2x+y} \end{bmatrix}$$

$$\Rightarrow Hf(0,0) = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

$$\frac{1}{2}\left[f_{xx}(0,0)(x-0)^2 + f_{xy}(0,0)(x-0)(y-0)_{(0,0)} + f_{yx}(0,0)(x-0)(y-0) + f_{yy}(y-0)^2\right]$$

$$P_2(x,y) = P_1(x,y) + \underbrace{\frac{1}{2}[x-0 \quad y-0]\, Hf(0,0)\begin{bmatrix} x-0 \\ y-0 \end{bmatrix}}$$

$$= P_1(x,y) + \frac{1}{2}[x \quad y]\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

$$= 1 + 2x + y + \frac{1}{2}(4x^2 + 2xy + 2xy + y^2)$$

$$= 1 + 2x + y + 2x^2 + 2xy + \frac{1}{2}y^2$$